

Ancillary Statistic (somewhat opposite concept of sufficient stat)

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Para Inf.

Suppose $\underline{X} = (X_1, \dots, X_n)$ has pmf/pdf $f(\underline{x}|\theta); \theta \in \Theta$
A statistic $A(\underline{X})$ is called "Ancillary for θ " if the distribution of $A(\underline{X})$ do not depend on θ .

Weaker notion of Ancillary: First order ancillary: $A(\underline{X})$ is a first order ancillary statistic for θ if $E_\theta[A(\underline{X})] = c$ (free of θ)

Any function of ancillary statistic is also ancillary.

Ex: ① $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$

$$a) A(\underline{X}) = \sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi^2_{n-1}$$

$$b) A(\underline{X}) = X_{(n)} - X_{(1)}$$

$$Z_i = X_i - \mu$$

$$= (Z_{(n)} + \mu) - (Z_{(1)} + \mu)$$

$$= Z_{(n)} - Z_{(1)}$$

② $X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta)$

$$X_i/\theta \sim U(0, 1)$$

$$A(\underline{X}) = \frac{X_1}{X_2} = \frac{X_1/\theta}{X_2/\theta}$$

③ $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

$$\theta = (\mu, \sigma^2)$$

Ancillary stat for θ

$$A_1(\underline{X}) = \frac{X_1 - X_2}{X_3 - X_4} \longrightarrow \frac{\frac{X_1 - \mu}{\sigma} - \frac{X_2 - \mu}{\sigma}}{\frac{X_3 - \mu}{\sigma} - \frac{X_4 - \mu}{\sigma}}$$

$$A_2(\underline{X}) = \frac{X_{(n)} - X_{(1)}}{S}$$

$$A_3(\underline{X}) = \frac{|X_1 - \bar{X}|}{S}$$

$$A_4(\underline{X}) = \frac{\sum (X_i - \bar{X})^3}{(\sum (X_i - \bar{X})^2)^{3/2}} \quad (\text{Skewness})$$

Location/Scale/Location-Scale Family

Suppose X has pdf $f(x|\theta)$, $\theta \in \Theta$. The family $\{f(x|\theta) : \theta \in \Theta\}$ is a

1) location family if $\theta \in \mathbb{R}$, $f(x|\theta) = g(x - \theta)$
 $X \stackrel{d}{=} \xi + \theta \longrightarrow \text{pdf } g(\cdot)$ where g is a known pdf (does not involve θ)
 $\xrightarrow{\text{location parameter}}$

2) scale family if $\theta > 0$, $f(x|\theta) = \frac{1}{\theta} g\left(\frac{x}{\theta}\right)$
 $X \stackrel{d}{=} \theta \xi \longrightarrow \text{pdf } g(\cdot)$ where g is a known pdf (does not involve θ)
 $\xrightarrow{\text{scale parameter}}$

3) location-scale family if $\theta_1 \in \mathbb{R}, \theta_2 > 0$ $\theta = (\theta_1, \theta_2)$
 $X \stackrel{d}{=} \theta_2 \xi + \theta_1$ $f(x|\theta) = \frac{1}{\theta_2} g\left(\frac{x - \theta_1}{\theta_2}\right)$ --

Ex: Location Family: $N(0,1)$, $\text{Cauchy}(0,1)$, $\text{Laplace}(\mu,1)$,

$$\left\{ \frac{1}{\pi \{1+(x-\theta)^2\}} \right\}$$

$$U(\theta, \theta+1), \text{Exp}(\theta, 1) \longrightarrow e^{-(x-\theta)} \mathbb{1}(x \geq \theta)$$

Scale Family: $N(0, \sigma^2)$, $\text{Cauchy}(0, \theta)$, $\text{Laplace}(0, \sigma)$,
 $U(0, \theta)$, $\text{Exp}(1, \theta)$

Location-Scale Family: $N(\mu, \sigma^2)$, $\text{Cauchy}(\theta, \sigma)$, $\text{Laplace}(\mu, \sigma)$,
 $\text{Exp}(\mu, \sigma)$

$$\downarrow$$

$$f(x|\mu, \sigma) = \frac{1}{\sigma} e^{-\frac{|x-\mu|}{\sigma}} \mathbb{1}(x \geq \mu)$$

$$\downarrow$$

$$f(x|\mu, \sigma) = \frac{1}{2\sigma} e^{-\frac{|x-\mu|}{\sigma}}$$

$$\text{Gamma}(\alpha, \beta) \longrightarrow f(x|\alpha, \beta) = \frac{e^{-x/\beta} x^{\alpha-1}}{\Gamma(\alpha) \beta^\alpha}$$

(Neither of three)

$\alpha = \text{Shape parameter}$

Def: A statistic $T(X)$ is called

1. location-invariant if $T(\underline{x} + c \underline{1}) = T(\underline{x}) \quad \forall \underline{x} \quad \forall c \in \mathbb{R}$
2. scale-invariant if $T(c \underline{x}) = T(\underline{x}) \quad \forall c > 0$
3. location-scale invariant if $T\left(\frac{\underline{x} - c_1 \underline{1}}{c_2}\right) = T(\underline{x}) \quad \forall \underline{x} \quad \forall c_1 \in \mathbb{R} \quad \forall c_2 > 0$

Ex: $X \sim f(x|\theta) = g(x-\theta) \quad T_1(X) = \frac{1}{n-1} \sum (X_i - \bar{X})^2,$

$$X \stackrel{d}{=} \bar{X} + \theta$$

$\uparrow$
pdf

$$X_{(n)} - X_{(1)}$$

location invariant

Ex. (Location-Scale Invariant) $T(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}$

Kurtosis $\frac{\sum (X_i - \bar{X})^4}{(\sum (X_i - \bar{X})^2)^2}$

Result: $X_1, \dots, X_n$ iid from $f(x|\theta)$ which is either location/scale/location-scale and if $T(X)$ is location/scale/location-scale invariant $\Rightarrow T(X)$ is ancillary for θ

Completeness

Suppose X has pmf/pdf $f(x|\theta); \theta \in \Theta$. The family $\mathcal{P} = \{f(x|\theta): \theta \in \Theta\}$ is called complete if for any real-valued $g(\cdot)$

$$E_{\theta}(g(X)) = 0 \quad \forall \theta \in \Theta \Rightarrow P_{\theta}(g(X) = 0) = 1 \quad \forall \theta \in \Theta$$

Ex 1. $\mathcal{P} = \{\text{Bin}(n, \theta): \theta \in (0, 1)\}$ complete

Let's consider any fn. g s.t. $E_{\theta}[g(X)] = 0 \quad \forall \theta \in (0, 1)$

$$\Rightarrow \sum_{x=0}^n g(x) \binom{n}{x} \theta^x (1-\theta)^{n-x} = 0 \quad \forall \theta \in (0, 1)$$

$$\Rightarrow \sum_{x=0}^n g(x) \binom{n}{x} \left(\frac{\theta}{1-\theta}\right)^x = 0 \quad \forall \theta \in (0, 1)$$

$$\Rightarrow \sum_{x=0}^n g(x) \binom{n}{x} \underbrace{\theta^x}_{\text{polynomial of degree } n} = 0 \quad \forall \theta > 0$$

cannot have more than n roots

$$\Rightarrow g(x) \binom{n}{x} = 0 \quad \forall x \in \{0, \dots, n\}$$

$$\Rightarrow g(x) = 0 \quad \forall x \in \{0, \dots, n\}$$

$$P_\theta(g(X) = 0) = P_\theta(X \in \{0, \dots, n\}) = 1$$

Ex 1. $\{N(0, \theta) : \theta > 0\}$ Take $g(X) = X$

$$E g(X) = 0 \quad \forall \theta > 0$$

$$\text{but } P_\theta(g(X) = 0) = 0 \quad \forall \theta > 0$$

Ex: Show $f(x|\theta) = \frac{1}{\theta}$ for $x=1, \dots, \theta$ $\theta \in \mathbb{N}$

$\mathcal{P} = \{f(x|\theta) : \theta \in \mathbb{N}\}$ complete. Then show $\mathcal{P}_0 = \{f(x|\theta) : \theta \in \mathbb{N} \setminus \{\theta_0\}\}$ is not complete (where θ_0 is any fixed natural no.)

Consider any function $g : E_\theta[g(X)] = 0 \quad \forall \theta \in \mathbb{N}$

$$\sum_{x=1}^{\theta} \frac{1}{\theta} g(x) = 0 \quad \forall \theta \in \mathbb{N}$$

$$\Rightarrow \sum_{x=1}^{\theta} g(x) = 0 \quad \forall \theta \in \mathbb{N}$$

$$\Rightarrow g(x) = 0 \quad \forall x \in \{1, 2, \dots, \theta\}$$

$$P_\theta[g(X) = 0] = 1 \quad \forall \theta \in \mathbb{N}$$

($\mathcal{P}_0$) For any g $E_\theta[g(X)] = 0 \quad \forall \theta \in \mathbb{N} \setminus \{\theta_0\}$

$$\Rightarrow \sum_{x=1}^{\theta} g(x) = 0 \quad \forall \theta \in \mathbb{N} \setminus \{\theta_0\}$$

$$\Rightarrow g(1)=0, g(2)=0, \dots, g(\theta_0-1)=0$$

$$\sum_{i=1}^{\theta_0+1} g(x)=0 \Rightarrow g(\theta_0)+g(\theta_0+1)=0$$

Take any g

$$g(x) = \begin{cases} 0 & \text{if } x=1, \dots, \theta_0-1, \theta_0+2 \\ c & \text{if } x=\theta_0 \\ \neq 0 & \\ -c & \text{if } x=\theta_0+1 \end{cases}$$

$$P_{\theta_0+1}[g(X)=0] = P_{\theta_0+1}[X \in \{1, \dots, \theta_0-1\}] = \frac{\theta_0-1}{\theta_0+1} \neq 1$$

Defⁿ: Suppose X has pmf/pdf $f(x/\theta) : \theta \in \Theta$. A statistic $T(X)$ is called complete if the family of pmf/pdf of T is complete i.e. in other words for any real valued fn. g if $E_{\theta}[g(T)] = 0 \quad \forall \theta \in \Theta$
 $\Rightarrow P_{\theta}(g(T)=0) = 1 \quad \forall \theta \in \Theta$

[A statistic T is complete if no non-constant fn of T is even first-order ancillary]

A statistic T is first-order ancillary $\Rightarrow$

$$E_{\theta}[T] = c \text{ (free of } \theta)$$

Qn: If T is complete and $U = h(T)$ for fn h , can we say U is complete

$$\begin{aligned}
&\text{Take any } g: \mathbb{E}_\theta[g(U)] = 0 \quad \forall \theta \\
&\Rightarrow \mathbb{E}_\theta[g(h(T))] = 0 \quad \forall \theta \\
&\quad \quad \quad \underbrace{g(h(T))}_{g^*(T)} \\
&\Rightarrow \mathbb{P}_\theta[g^*(T) = 0] = 1 \quad \forall \theta \\
&\Rightarrow \mathbb{P}_\theta[g(U) = 0] = 1 \quad \forall \theta
\end{aligned}$$

Q: Completeness $\Rightarrow$ Sufficiency? No
 Sufficiency $\Rightarrow$ Completeness? No